BACHELOR OF COMPUTER APPLICATIONS | D Y PATIL INTERNATIONAL UNIVERSITY

SMART CIVICFIX

CLOUD-BASED CIVIC COMPLAINT MANAGEMENT

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01 | INTRODUCTION

The Problem: Urban India's Civic Crisis

Urban India faces a deepening civic infrastructure crisis. Traditional complaint mechanisms — phone calls, handwritten applications, and in-person visits — are slow, fragmented, and nearly impossible to track.

Citizens lack transparent channels to report potholes, garbage, water leakages, and broken streetlights, leading to delays, poor accountability, and eroding public trust.

500M+

Urban residents affected by civic infrastructure issues

No SLA

Most cities have no defined resolution deadlines for complaints

Offline

Majority of citizens still rely on manual, offline complaint systems

02 | OBJECTIVES

What Smart CivicFix Aims to Achieve

1

Digital Reporting Platform

Simple web-based interface for reporting potholes, garbage, water leaks, and streetlight issues.

2

Automatic Location Capture

GPS-based location detection captured automatically at the time of complaint submission.

3

Smart Department Assignment

Auto-routing of complaints to the correct municipal department without manual intervention.

4

SLA-Based Resolution

Defined deadlines for each civic issue category to ensure accountability and timely action.

5

Complaint Tracking & Transparency

Real-time status visibility from submission through to final closure.

6

Citizen Reward System

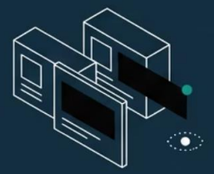
Reward points for valid complaints to encourage active civic participation.

03 | GAP IDENTIFICATION

Where Existing Systems Fall Short

Current civic complaint systems across Indian cities share critical weaknesses that prevent effective urban governance. Citizens are left without visibility, authorities lack actionable data, and no scalable solution exists for multi-city deployment.

Phone calls and handwritten submissions cause massive delays



No Transparency. Citizens cannot track complaint progress, trust erodes

Not Scalable. City-specific apps cannot be replicated across regions

No Analytics.

No Analytics.

Authorities **lack dashboards to monitor trends or performance**

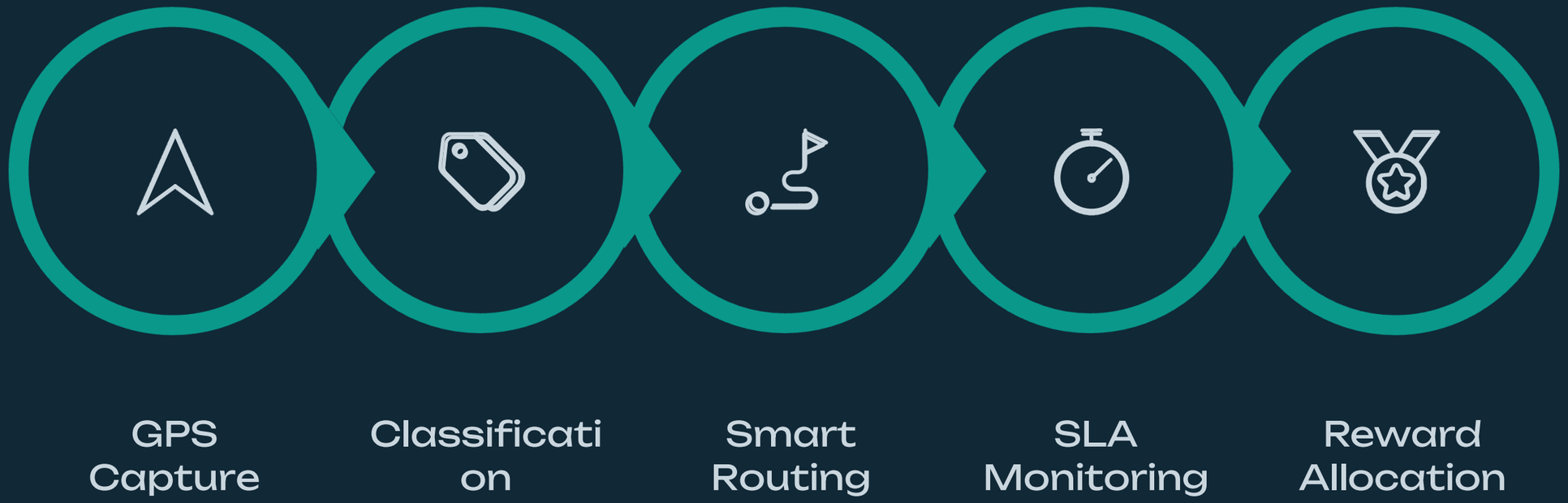
No Cloud Integration. Existing systems lack secure, scalable cloud infrastructure

Limited Scope. Most digital tools address only only one issue type, no unified platform exists

Exisiting Systems Review

App/System	What it does	Limitation
🧹 Swachhata App	Reports sanitation issues under Swachh Bharat Mission	Only covers sanitation, no road/water/electricity
📞 Indore 311	Citizens report road damage, drainage via mobile	City-specific, not scalable
🏙️ BBMP Sahaya	Bangalore civic complaint portal	No SLA tracking, poor transparency
📝 MyGov Portal	General citizen grievance platform	Not real-time, slow resolution

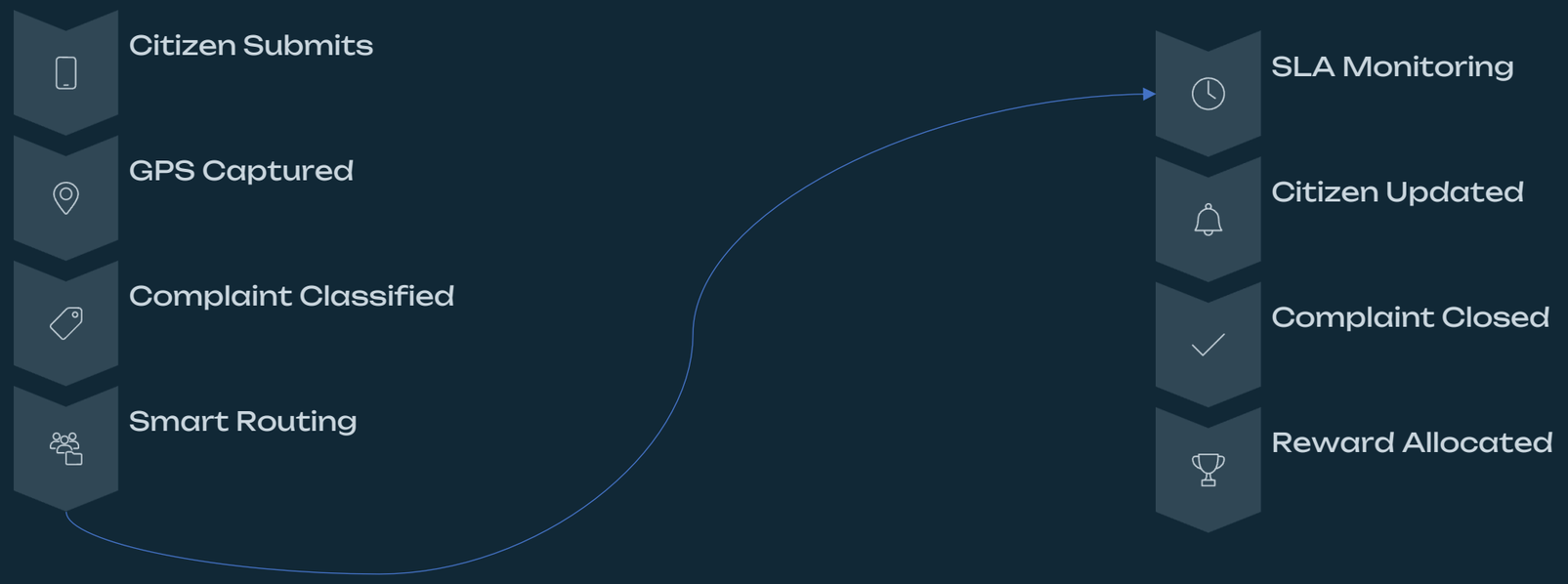
Core Algorithms Powering Smart CivicFix



Each algorithmic step operates automatically upon complaint submission, ensuring zero manual overhead for citizens while maintaining full accountability across municipal departments.

05 | SYSTEM WORKFLOW

How a Complaint Flows Through Smart CivicFix



End-to-End Accountability: From the moment a citizen submits a complaint with GPS coordinates, the system automatically classifies, routes it to the correct department, and begins the SLA countdown. Real-time notifications keep citizens informed at every stage, and reward points are allocated upon closure — creating a fully transparent civic management loop.

06 | TECHNOLOGY STACK

Platform Architecture

Frontend (Citizen & Admin)

React.js / HTML5 · GPS API Integration · Real-time Status Dashboard · Responsive Web Interface

Backend & Logic

Node.js / Python (Flask) · Complaint Classification Engine · Smart Routing Algorithm · SLA Monitoring & Escalation

Cloud & Database

AWS / Firebase · NoSQL / SQL Database · Secure Complaint Storage · Scalable Multi-city Deployment

☁️ Cloud-Native | 📍 GPS-Enabled | 🤖 AI-Assisted Classification | 🏢 Multi-Department Integration | 🏆 Citizen Engagement

07 | PROJECT TIMELINE

8-Week Development Roadmap

1 — **Week 1**
Requirement Analysis

2 — **Week 2**
System Design

3 — **Week 3**
Algorithm Development

4 — **Week 4**
Platform Development

5 — **Week 5**
Testing & Debugging

6 — **Week 6**
Deployment

7 — **Week 7**
Feedback

8 — **Week 8**
Optimization

A structured 8-week sprint ensures disciplined delivery — from initial requirements gathering through to a fully deployed, tested, and optimised civic complaint management platform ready for pilot deployment.

08 | EXPECTED OUTCOMES

Impact of Smart CivicFix

Faster Resolution

SLA-driven deadlines ensure complaints are resolved within committed timeframes, significantly reducing backlog.

Full Transparency

Real-time tracking gives citizens complete visibility into each stage of their complaint's lifecycle.

Better Governance

Admin dashboards help municipalities monitor trends, measure departmental performance, and plan proactively.

Scalable Citywide

Cloud-based architecture allows rollout across multiple cities without rebuilding the system from scratch.

Civic Engagement

Reward points incentivise citizens to actively report issues, driving platform adoption and participation.

Data-Driven Decisions

Analytics on complaint frequency and resolution patterns empower smarter urban planning and resource allocation.

Thank You

Building Transparent, Accountable Urban Governance

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